



Arun K

Alphonse J S
Niranjan J
Gokul S
Gautham Krishna
T.Maheshwaran
Rohan Tom Reji
Abhinav Manoj
Rithu Deep
Abhiram Vijay kumar
Alan Francis

Project Title: Sustainable AI-Based Smart Home and Autonomous Robot System

Objective:

To develop a smart room replica that manages energy-efficient operations (light, fan, AC, and window control) and an autonomous robot car that monitors environmental conditions, focusing on temperature, humidity, and detecting harmful gases like carbon monoxide. The system aims to enhance safety and energy efficiency through real-time automation and alerts via a Telegram bot.

System Components:

1. Room Replica:

- **LED Light:** Represents the ceiling light, controlled based on time and motion detection.
- **Fan (300 RPM Motor):** Simulates a ceiling fan, controlled by room temperature.
- **AC Replica:** A small fan controlled by the ESP32, simulating the air conditioning system.
- **Servo Motor:** Controls the window, opening or closing it based on temperature.
- **MPU6050 (Accelerometer/Gyroscope):** Placed outside the room to measure the outside temperature (not for motion tracking).

2. Autonomous Robot Car:

- **Ultrasonic Sensor:** Detects obstacles, enabling autonomous navigation.
 - **PIR Sensor:** Detects motion around the car.
 - **Smoke and Carbon Monoxide Detector:** Detects smoke and harmful gases, particularly carbon monoxide, with alerts sent via the Telegram bot if carbon monoxide is detected.
 - **DHT11 Sensor:** Monitors temperature and humidity around the car.
 - **Motor Driver (L298N or similar):** Controls the robot car's wheels for autonomous movement.
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System Workflows:

Room Control:

- **Light and Fan:** The light and fan are managed based on motion detection and room temperature to save energy.
- **AC Replica and Window Control:** The AC is activated based on temperature, and the window is automatically controlled using a servo motor.

Robot Car:

- **Autonomous Navigation:** The car uses an ultrasonic sensor to navigate around obstacles and a PIR sensor to detect motion.
- **Environmental Monitoring:** The DHT11 sensor records temperature and humidity around the car. The smoke and CO detector monitors harmful gases, sending alerts to warn of potential dangers, especially from carbon monoxide.

ESP32 Communication and Alerts:

- **Master-Receiver Protocol:** The ESP32s in the room replica and robot car communicate with each other for data transfer.
- **Telegram Alerts:** The system sends real-time updates, including:
 - Outside temperature and motion alerts.
 - Smoke or carbon monoxide detection warnings.
 - Window control status based on temperature thresholds.
 - Energy-saving tips and updates related to sustainable practices.